

Utsav Pareshkumar Soni

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PROFILE

Dynamic and passionate AI Engineer with expertise in Generative AI, Vision-Language Models, and the creation of custom AI systems. Adept at designing, developing, and deploying advanced AI applications to address real-world challenges, including accessibility solutions and retrieval-augmented generation (RAG) systems. Proficient in utilizing state-of-the-art AI frameworks, cloud platforms, and open-source tools to deliver innovative and impactful solutions.

EDUCATION

Bachelor of Engineering in Electrical Engineering, Adani Institute of Infrastructure Engineering 08/2017 – 06/2021
- CGPA 9.24 Ahmedabad, Gujarat

PROFESSIONAL EXPERIENCE

Associate Software Engineer - IoT, Rishabh Software 10/2023 – present
Vadodara, Gujarat

- Engineered a Wi-Fi-enabled smart machine, enabling real-time data collection and creating a foundation for AI-powered predictive analytics and automation.
- Designed a real-time monitoring pipeline with Grafana and JSON-based data flows, skills transferable to building dynamic dashboards for AI insights.
- Built secure IoT-cloud communication pipelines using MQTT, aligning with data preprocessing needs for AI models and Generative AI systems.
- Delivered scalable Azure-based solutions, integrating device provisioning and cloud storage, with the potential for AI model hosting and deployment

Assistant System Engineer, Tata Consultancy Services 07/2021 – 09/2023
Ahmedabad, Gujarat

- Translated technical requirements into actionable solutions, building strong client relationships and ensuring the successful delivery of complex projects.
- Managed client databases, ensuring data integrity, security, and optimal performance to support seamless system operations.
- Led a team of test engineers, ensuring efficient task execution and maintaining high-quality standards throughout the project lifecycle.

PROJECTS

Advanced AI Vision Assistant, (Github: <https://github.com/utsav-soni/iksika.i>)

- Pioneered the development of a groundbreaking vision AI application utilizing Meta's LLaMA 3.2 Vision LLM, enabling real-time scene analysis and multilingual text-to-speech functionality across 100+ languages.
- Achieved an impressive P50 latency of 3.17s using Groq's Inference Engine, delivering instant descriptions for live scenes and uploaded images via a user-friendly Streamlit interactive app.
- Designed a modular architecture, integrating advanced LangSmith for traceable insights, Deep Translator for language versatility, and Google Text To Speech (gTTS) for seamless audio feedback, ensuring scalability for tourism, education, and healthcare applications.

Custom Retrieval-Augmented Generation (RAG) System

- Built a fully custom RAG system from the ground up vector database, and state-of-the-art Llama3-8b-8192 model, showcasing expertise in personalized dataset integration and Generative AI workflows.
- The system was deployed on AWS EC2 (Free Tier) using Docker, optimized for efficient scaling with ChromaDB as a high-performance vector store for efficient data retrieval in the RAG system.
- Integrated LangSmith for robust debugging, testing using various prompt engineering techniques, and monitoring, delivering real-time responses through an intuitive Streamlit interface.
- API-enabled the system using LangServe, making it integration-ready for dynamic enterprise-level applications, including knowledge retrieval and interactive AI assistants.

SKILLS

Generative AI Frameworks: Hugging Face | LangChain | crewAI | Ollama | Meta LLaMA models | Amazon SageMaker and Bedrock

Deployment and Cloud Platform: AWS | Azure | Docker | Streamlit | Grafana | Firebase | Supabase

Programming Language: Python | C

Data Handling, Processing and Collaboration: Optical Character Recognition (OCR) | Natural Language Processing (NLP) | Computer Vision Techniques using OpenCV | PyTorch | VScode | Google Colab | Jupyter Notebook | Git Version Control

ACHIEVEMENTS

Certification: Generative AI with Large Language Models | DeepLearning.AI 🏆

ABU Robocon 2018-2020, IIT Delhi

- Achieved AIR 2 at an International-level Robotics Competition, showcasing teamwork, and problem-solving skills.